Department of Electrical Engineering

विद्युत अभियांत्रिकी विभाग

Electronics Lab

इलेक्ट्रॉनिक्स लैब

Experiment Number 7

प्रयोग संख्या 7

Study of Wein Bridge Oscillator.

वेन ब्रिज ऑसिलेटर का अध्ययन।

MAULANA AZAD NATIONAL INSTITUTEOF TECHNOLOGY, BHOPAL

ELECTRONICS LAB

EXPERIMENT NO.7

Aim : Study of Wein's bridge oscillator

Instruments Required:

- Fixed Output DC Regulated Power Supply of ±15 Volts.
- IC 741 is placed inside the cabinet & important connections are brought out on sockets.
- One Potentiometer is mounted on the front panel to vary the amplitude of output signal
- Three set of Resistances (R) and Capacitors (C) for tuned circuit are also provided on the front panel
- Circuit diagram is printed on the front panel and respective components are also given on the front panel.

Circuit Diagram :

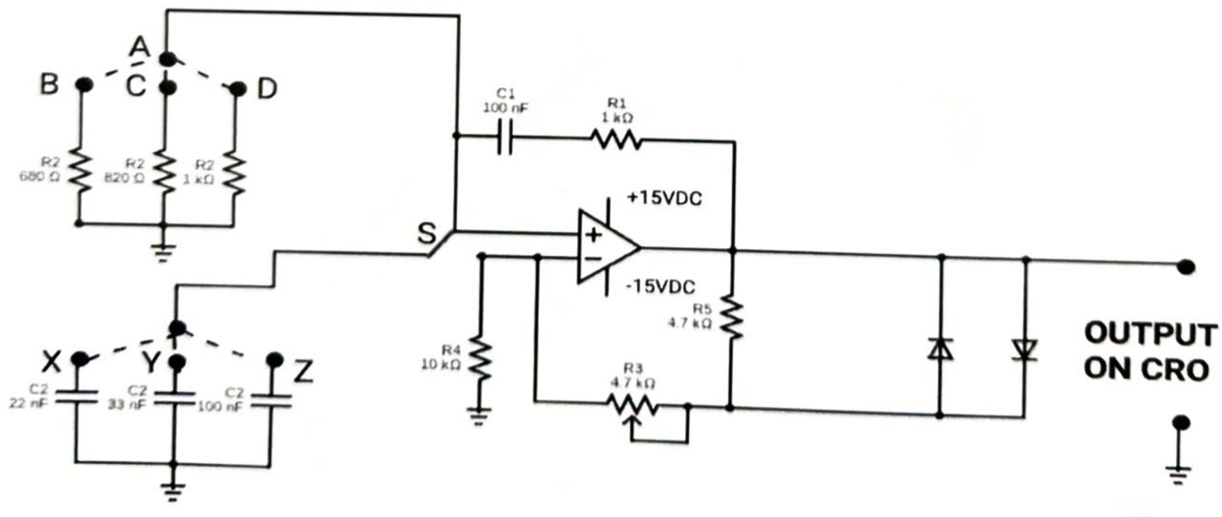

Fig 1 : Circuit diagram for study of Wein's bridge oscillator

Theory :

The Wein Bridge Oscillator is a type of electronic oscillator that generates sinusoidal waveforms at its output. The circuit consists of a feedback network of resistors and capacitors, along with an amplifier, which provides gain to the circuit. The circuit operates on the principle of positive feedback, in which a portion of the output voltage is fed back to the input of the amplifier in phase with the input signal.

Oscillator is an important device for many electronic circuit applications and its prime function is to generate waveforms at constant amplitude and desired frequency. Basically an oscillator is an electronic circuit which converts DC supply voltage to an output waveform of some frequency. The oscillator circuit must also be capable of producing sustained oscillations. The oscillators are classified into two basic categories: Sinusoidal & Non-sinusoidal. If the waveform generated looks like a sine wave, the circuit is called a sinusoidal oscillator and the circuit producing all other waveforms is called a non-sinusoidal oscillator. Sometimes, the oscillators are also classified on the basis of frequency of the generated waveform, viz. Audio frequency. radio frequency and ultra high frequency oscillators.

The feedback network in the Wein Bridge Oscillator consists of two RC circuits, each with a resistor and a capacitor in series. The two RC circuits are connected in a bridge configuration, where the output of one RC circuit is connected to the input of the other RC circuit, and vice versa. This bridge configuration results in a balanced circuit, where the impedance of one branch is equal to the impedance of the other branch.

Each oscillator has a tank. This tank circuit consists of an Inductance coil (L) or Resistance (R) connected in parallel with Capacitor (C). The frequency of oscillations in the circuit depends upon the value of the coil or resistance and capacitance of the capacitor. The frequency of the oscillation is determined by the values of the Capacitor & Inductor.

The circuit of Wein Bridge Oscillator is drawn on the front panel of the instrument. The oscillator consists of a tuned circuit and a feedback network. For obtaining constan output negative feedback is introduced in the circuit.

Procedure :

1. The circuit of Wein Bridge O cillator is drawn on the front panel of the instrument. The cillator consists of a tuned circuit and a feedback network. For obtaining constant output negative feedback is introduced in the circuit.
2. Connect any one value of R2 & C2 in the circuit at Pin No. 3 of IC 741 by connecting dotted lines through patch cords.
3. Switch "ON" the instrument using ON/OFF toggle switch provided on the front panel. Also switch "ON" the CRO.
4. Connect the output terminal of the operational amplifier to the oscilloscope.
5. Connect the function generator to the input of the Wein Bridge oscillator circuit.
6. Observe the output waveform on CRO & vary the amplitude of the signal using potentiometer (R3) provided on the front panel.

7. Adjust the frequency of the function generator until the output waveform becomes sinusoidal.

8. Observe and record the frequency and amplitude of the output waveform.
9. Now note down the frequency of oscillations & formula used to calculate the frequency to

$$f = \frac{1}{(2\pi(R1 * C1 * R2 * C2)^{1/2})}$$

10. Repeat the experiment for other values of Capac tors (C2) & Resistance (R2)

Observations:

Record the frequency of the output waveform for each s t of resistors and capacitors used

Sr. No	R2 (Ω)	C2 (F)	Theoretical Frequency (KHz)	Practical Freque cy (KHz)	E ror (%)

Conclusion :

The Wein Bridge oscillator circuit is a simple and effective circuit for generating sinusoidal waveforms. By changing the values of resistors and capacitors, the frequency and amplitude of the output waveform can be adjusted. The circuit can be further improved by using more precise components and adding a buffer stage to isolate the oscillator from the load.

Result :

Questions:

1. What is a Wein's Bridge Oscillator and how does it work?
2. What is the frequency of oscillation in a Wein's Bridge Oscillator?
3. What are the common applications of a Wein's Bridge Oscillator?
4. What are the limitations of a Wein's Bridge Oscillator?
5. What is the condition for sustained oscillations in the Wein's Bridge Oscillator?